

Koa Shen

Mechanical Engineering Candidate · Hardware Design · Analysis · Manufacturing

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EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering · 3.97 major / 3.94 overall GPA · Dean's Honors every quarter

Santa Barbara, CA

- Coursework: Dynamics, Mechanics of Materials, Circuits. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Built CAD-import, kinematics, collision-check, and load-case workflows for high-DOF precision motion assemblies.
- Redesigned a high-DOF X-ray optics assembly with new mounts and stages; produced GD&T-controlled Solid Edge designs and Teamcenter documentation.

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Maintained 100+ interactive exhibits; reverse-engineered and fabricated mechanisms, leading three overhauls and avoiding \$3,000+ in replacement costs.

Fabrication Team Lead

Fall 2024 – Present

Gaucho Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- Led 15 people through safety-system design, fabrication, and installation; passed FSAE technical inspection.
- CNC-machined 33 steel and aluminum components, holding bearing interfaces within 0.0005 in.

SELECTED PROJECTS

Digital Twin Simulation Framework for LCLS Motion Stages

2026 – Present

- Built a Python/Drake digital twin for a 32-DOF assembly, integrating CAD import, kinematics, collision checking, and MeshCat visualization.
- Applied the framework to three LCLS assemblies; collision checks run in milliseconds across 305 bodies.

Polycapillary Optics Assembly Redesign

2026 – Present

- Redesigned a 32-DOF optics assembly with three kinematic bases, six hot-swappable optics, and added stages to improve alignment and packaging.
- Used simulation findings and statics/dynamics calculations to guide design changes and evaluate handling loads.

Compact Cycloidal Reducer & Stepper Package

Summer 2026 – Present

- Built and bench-validated closed-loop stepper control for a 15:1 cycloidal reducer using custom C++ firmware and AS5600 encoder feedback.
- Implemented missed-step detection and automatic position-error compensation; multi-joint integration remains in development.

FSAE Steering Wheel Development

Fall 2024 – Spring 2026

- Integrated controls, display, and quick-release into a competition steering wheel; FEA-guided iteration reduced mass from 5 lb to 1.7 lb.

TECHNICAL SKILLS

Mechanical Design & Manufacturing: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, CNC/manual machining, Mastercam

Analysis & Simulation: Statics and dynamics, Drake kinematics, Collision detection, FEA-guided design, Python, MATLAB

Controls & Prototyping: C/C++, PlatformIO, Stepper control, AS5600 encoder feedback, Arduino, Git/GitHub